Junghoon Chung

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SUMMARY

PhD candidate in Industrial & Systems Engineering with a strong background in UX research, human factors, and data-driven design. Experienced in designing and executing end-to-end mixed-methods research including interviews, surveys, controlled experiments, and field studies. Skilled at integrating qualitative insights with large-scale behavioral and sensor data to inform product and system design. Proven ability to translate complex findings into actionable recommendations for cross-functional teams. Seeking a UX Research internship to support evidence-driven product development at scale.

EDUCATION

University of Washington, Seattle — Sep 2025 – Jun 2029 (Expected)
Ph.D in Industrial and Systems Engineering, Dean's Fellowship Recipient

Virginia Tech, Blacksburg — Aug 2023 – May 2025
M.S in Industrial and Systems Engineering (Thesis Track), GPA 3.89/4.0

Yonsei University, Seoul — Mar 2017 – Feb 2023
B.S in Industrial Engineering, GPA 3.52/4.0, Student Council President

EXPERIENCE

Graduate Research Assistant, Human Automation Systems Lab, University of Washington — Sep 2025 – Present

- Designed controlled experiments and user studies to evaluate attention, workload, and performance in complex task environments
- Collected multimodal behavioral and physiological data to understand user states during sustained interaction, and generated insights for systems that support sustained attention and reduce cognitive overload

Graduate Research Assistant, Occupational Ergonomics and Biomechanics Lab, Virginia Tech — Aug 2023 – May 2025

- Led end-to-end mixed-methods research on how knowledge workers interact with computer systems and workplace technologies, conducting interviews, surveys, and behavioral observations alongside sensor-based data collection
- Synthesized qualitative feedback and quantitative data to identify usability barriers and contextual drivers of behavior, and analyzed longitudinal data to understand patterns in productivity, comfort, and task engagement

KEY PROJECTS

Human Vigilance Modeling with Causal Inference | *University of Washington*
Investigated how cognitive load and alertness influence performance during sustained tasks by integrating behavioral measures with 30+ physiological features across 10,000+ observations per participant. Examined how multimodal signals relate to sustained attention via task reaction time, and compared model performance across time-series aggregation windows.

Driver Response to V2X Pedestrian Alerts | *City of Bellevue / University of Washington*
Designing a field study protocol with the City of Bellevue, T-Mobile, and Applied Information to understand how V2X pedestrian alerts shape driver awareness and yielding decisions in naturalistic, on-road conditions. Study includes IRB documentation, recruitment protocols, NASA-TLX workload surveys, and pre/post driving interviews. Data collection is planned for October.

Multi-Stage Behavioral Framework for Driver Takeover | *University of Washington*
Proposing a behavioral framework that decomposes automated-driving takeover into three interpretable stages (Attention Reorientation, Control Acquisition, Control Stabilization), aiming to diagnose where in the takeover process successful and unsuccessful drivers actually differ rather than only score overall performance.

NudgeWatch, Adaptive Sit-Stand Nudging System | *University of Washington*
Designed and built the full research instrument myself, a native Apple Watch app with a live posture-sensing engine, to test a context-aware nudging hypothesis no existing tool could support. Ran a 5-participant mixed-methods pilot, and found the adaptive loop measurably worked, but was rarely perceived correctly by users, a key insight for designing trustworthy adaptive systems. Portfolio: Junghoon-Chung.github.io/#nudgewatch

Context-Aware Sit-Stand Intervention | *Virginia Tech*
Conducted a longitudinal field study using 3 weeks of behavioral logs per participant (100,000+ records) to understand posture-switching behavior. Applied time-series forecasting and user clustering to identify contextual triggers of behavior change, informing adaptive workplace interventions that balance productivity and health.

SKILLS

Programming: Python, SQL (MySQL), Swift (WatchOS)
Research Methods: Interviews, Surveys, Usability Testing, Experimental Design
Analysis: Statistical analysis, Evidence synthesis
Collaboration: Cross-functional teamwork, Technical communication, Presentation

AWARDS

Dean's Fellowship, College of Engineering	University of Washington, 2025–2029
Social-Innovation Capability Video Contest 2nd Place, Data mining	Yonsei University, 2022
Merit-based Scholarship (Academic Excellence)	Yonsei University, 2021